The R-36M is already being outmoded by R&D in long-range missiles by countries like the U.S.

that of "Fat Man" - the bomb which was dropped on Nagasaki in August 1945, which means that a single warhead on a single missile is capable of generating over 5 million human casualties at its point of impact in a country like India.

The frightening nature of this destruction makes space all the more alluring for a war destination. And to that end the innovations in warfare have progressed with the same belief. Technologies that can support the advantages of a space assault on terrestrial nations as well as the potential for minimising collateral damage, outer space may in fact be the best place for war. And it presents its own set of challenges.

War Is All Kinds Of THEL

Early on during the Cold War military strategists realized that nuclear weapons in space were not a good idea. The nuclear fall out combined with the unpredictable debris showers towards the surface of the Earth would leave every aggressor nation at risk of unsuspecting damage. To compensate for this side-effect more powerful conventional weaponry or new innovations in weapons design would be needed. The foremost of these designs for space based weapon inevitably came from the imagination of science fiction - lasers, Advanced Tactical Lasers to be more precise.

An invention of the Boeing Company, the Advanced Tactical Laser system was commissioned by the United States Special Operations Command in 2002. Within six years the Boeing Company delivered the weapons strapped to a C-130H Hercules transport aircraft, and proved capable of defeating a ground based target by the end of 2009. The silent and invisible laser possess a 100 kilowatt capacity sufficient to slice a tank shell with ease. Other similar direct energy weapon in service and development are the High Energy Liquid Area Defense System, Tactical High Energy Laser (THEL) or the Nautilus laser system and the Mobile-THEL have all proven successful in neutralizing incoming mobile targets such as rockets and artillery shells.

If we theorise their use in outer space we can easily see how their strengths would play off against their disadvantages. The notorious over-heating of laser weapons is only limited by their heat-transfer support systems which would allow them to stay cooled. It is easy to imagine a space-based improvisation that would use the minus 270.45 Celsius temperature of space to cool the weapons allowing for near continuous usage. The Mobile-THEL laser system is designed for quick bursts of attacks to minimise its heating but in the sub-zero temperatures of space it could potentially operate continuously.

Velocitas Eradico - I, Speed, Destroyer of Worlds

An alternative to the complicated and difficult to construct direct energy weapons is the far simpler development of non-combustible high velocity inert projectile launchers - railguns. The simplicity of the railgun concept is built into its physics, which is the acceleration of projectiles at supersonic velocities using electromagnetic propulsion. By rapidly accelerating a projectile to supersonic speeds we no longer depend on chemical propellants like gunpowder which wouldn't perform effectively in the vacuum of space.

The MIRV weapon is seen executed in TV shows like Battlestar Galactica to great accuracy.

The current design of railguns only requires a power supply of pulsed direct current and a heat resistant material in its armature build. With these components military railguns do not even need explosive rounds as the kinetic energy of these projectiles is sufficient to cause considerable damage. The first weaponized railgun is already in production and is known as the General Atomics Blitzer system. This weapon launches discarded round casings of conventional artillery at velocities of Mach 5 (1600m/sec). The power and range of this weapon is proven to be significant as in tests the projectile continued to travel nearly 7 kilometres after having penetrated its 1/8 inch steel plate target.

In another test by the United States Navy in 2008, a 3.2kg projectile was launched at velocities of nearly Mach 7 producing 9.2 million joules of kinetic energy worth of damage. By using the kinetic energy of such projectile weapons in the relative gravity-free arena of space the potential for damage is equivalent to TNT explosive charges of the same mass. And as these types of weapons achieve greater velocity, their destructive capability becomes all the more greater.

Starfighters of the Future

No discussion of space war can be complete without talking about warships. Once it was understood that space in the next frontier, nations the world over have developed their own space vehicles. Fortunately, the development of spacecrafts has been driven by scientific and exploratory ideals. But with the creation of the X-37B Orbital Test Vehicle (OTV) this fact might very well change.

The weaponized laser already in production has surprising range and power for its size, promising more in future versions.

The Boeing X-37 series of spacecrafts are designed primarily to serve as reusable unmanned space vehicles intended to replace the Space Shuttle as the primary means of carrying out orbital activities. The X-37B va-